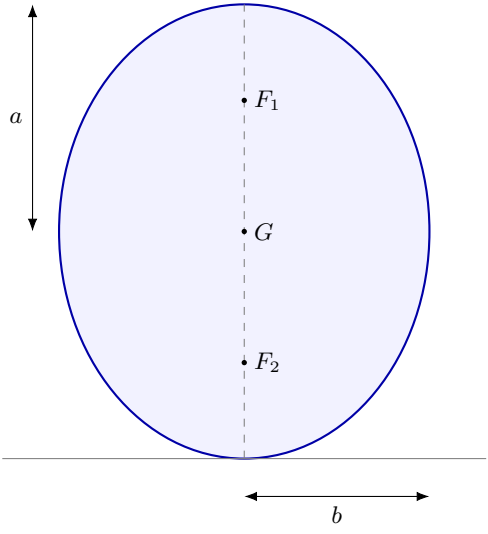
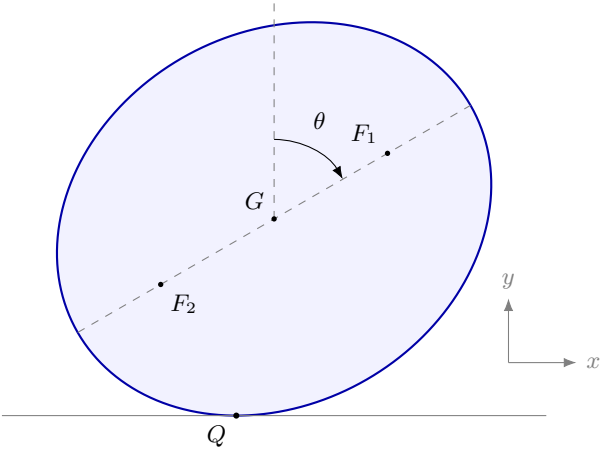
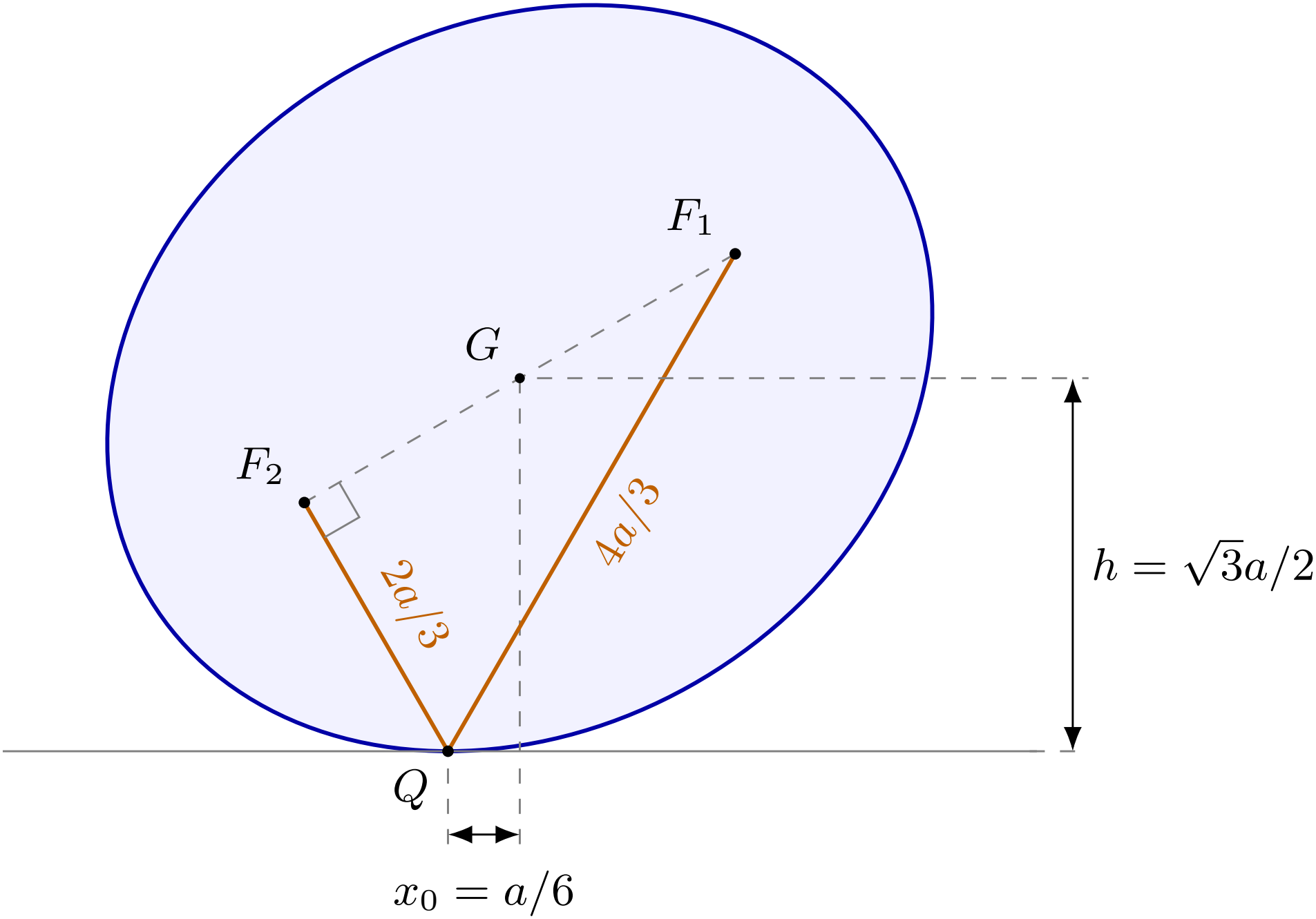


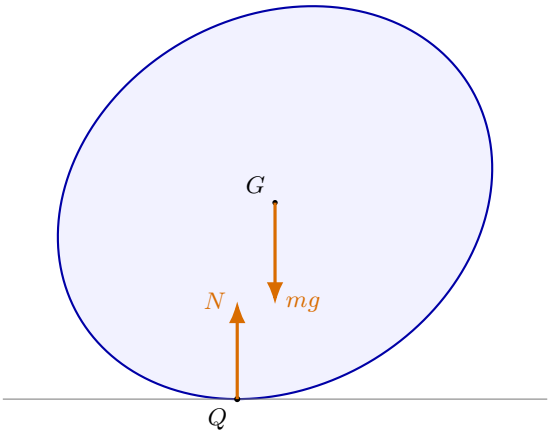
初始长轴竖直



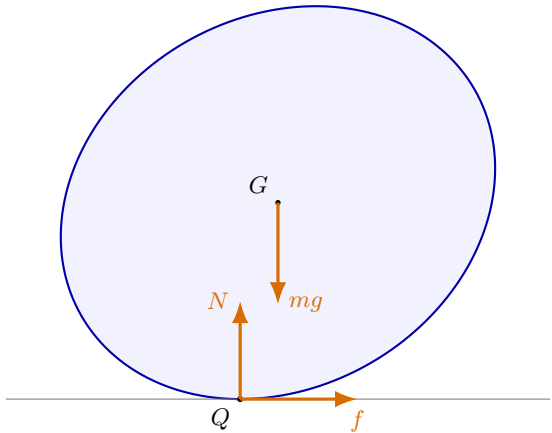
向右倾倒至 $\theta = \pi/3$


F_1
G
F_2
$2a/3$
$4a/3$
$h = \sqrt{3}a/2$
Q
$x_0 = a/6$

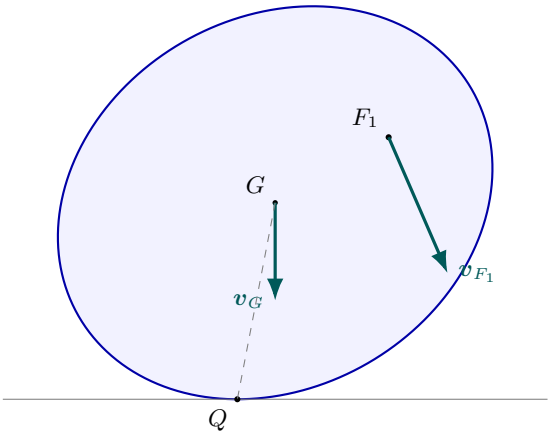
光滑接触：$f = 0$



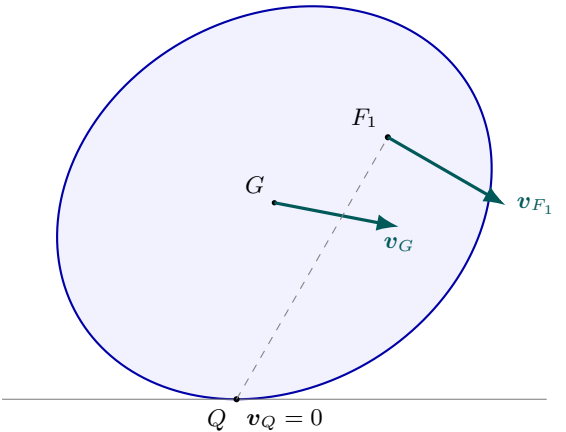
纯滚动：该时刻 $f > 0$

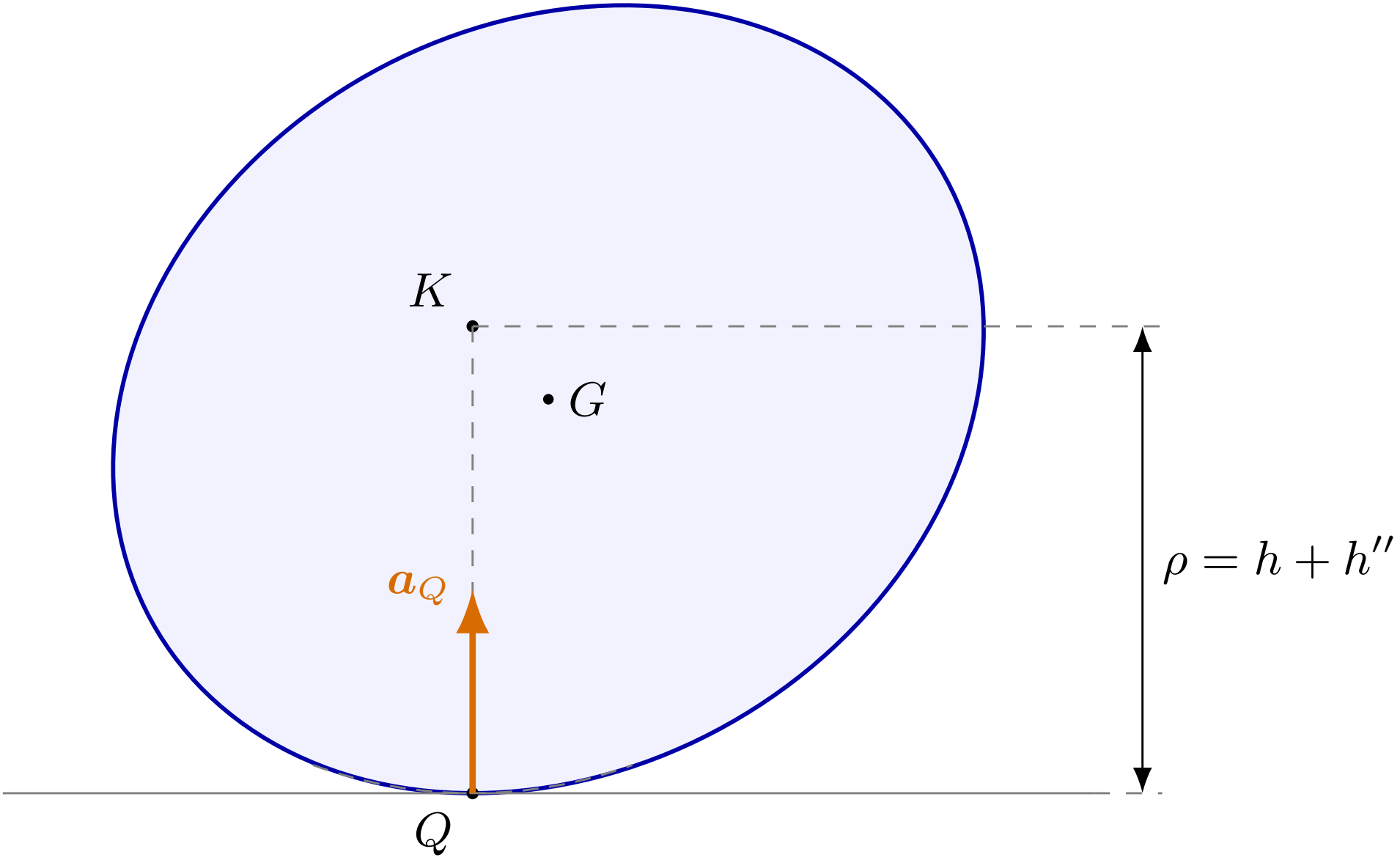


光滑接触：$\dot{X}=0$



纯滚动：$\dot{X}=h\omega$





$$\boldsymbol{v}_Q = 0, \quad \boldsymbol{a}_Q = \rho\omega^2 \boldsymbol{e}_y$$